

COLLEGE OF BUSINESS EDUCATION



COURSE: BACHELOR OF INFORMATION TECHNOLOGY

FACILITATOR: JOSEPH HAULE

MODULE NAME: COMPUTER NETWORK

MODULE CODE: ITU07209

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REG NUMBER: 03.4512.02.01.2024

YEAR OF STUDY: I

SEMESTER: 2nd

STREAM TNC

Topology

Topology requirements

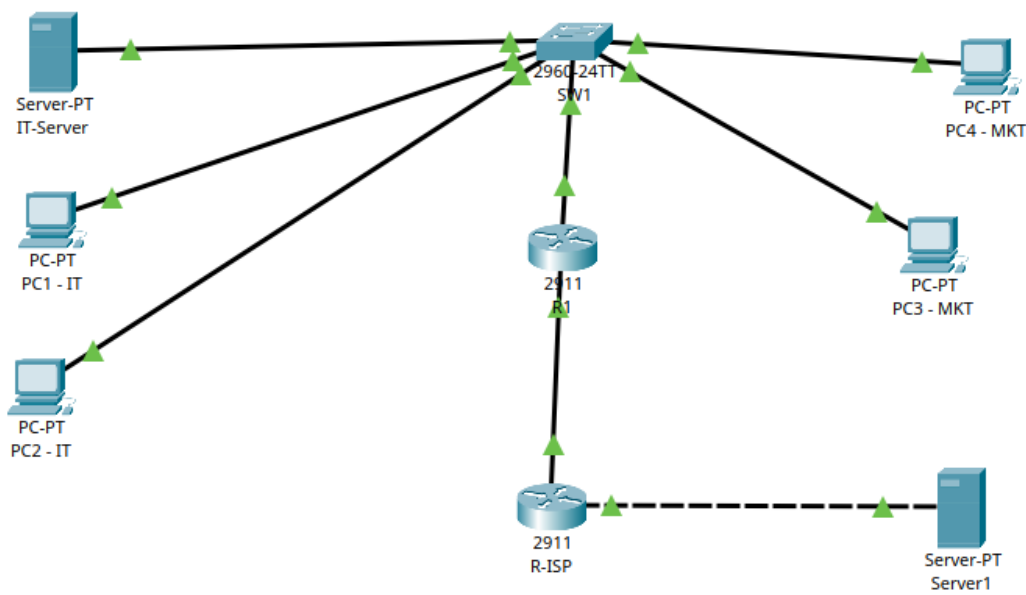
IP planning

Segment	Network	Gateway
VLAN 10 = IT	192.168.10.0/24	192.168.10.1
VLAN 20 = Marketing	192.168.20.0/24	192.168.20.1
WAN link R1 to R-ISP	200.200.200.0/30	R1=.1 / R-ISP=.2
Internet server	8.8.8.0/24	8.8.8.1 (on R-ISP)

End Devices IP planning

DNS = 8.8.8.8

Devices	IP address	Mask	Gateway
PC1	192.168.10.10	/24	192.168.10.1
PC2	192.168.10.11	/24	192.168.10.1
Server0	192.168.10.20	/24	192.168.10.1
PC3	192.168.20.10	/24	192.168.20.1
PC4	192.168.20.11	/24	192.168.20.1



Switch (SW1) Configuration

```

Switch#
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW1
SW1(config)#
SW1(config)#vlan 10
SW1(config-vlan)#name IT
SW1(config-vlan)#exit
SW1(config)#vlan 20
SW1(config-vlan)#name MKT
SW1(config-vlan)#exit
SW1(config)#
SW1(config)#interface range fa0/1-3
SW1(config-if-range)#switchport mode access
SW1(config-if-range)#switchport access vlan 10
SW1(config-if-range)#exit
SW1(config)#interface range fa0/4-5
SW1(config-if-range)#switchport mode access
SW1(config-if-range)#switchport access vlan 20
SW1(config-if-range)#exit
SW1(config)#
SW1(config)#interface gi0/1
SW1(config-if)#switchport mode trunk
SW1(config-if)#switchport trunk allowed vlan 10,20
SW1(config-if)#exit
SW1(config)#
SW1(config)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
SW1#
  
```

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Router (R1) Configuration

```
Router#enable
Router#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
R1(config)#interface gi0/0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
R1(config)#interface gi0/0.10
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
encapsulation dot1Q 10
R1(config-subif)#ip address 192.168.10.1 255.255.255.0
R1(config-subif)#ip nat inside
R1(config-subif)#exit
R1(config)#
R1(config)#interface gi0/0.20
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
encapsulation dot1Q 20
R1(config-subif)#ip address 192.168.20.1 255.255.255.0
R1(config-subif)#ip nat inside
R1(config-subif)#exit
R1(config)#
R1(config)#interface gi0/1
R1(config-if)#ip address 200.200.200.1 255.255.255.252
R1(config-if)#ip nat outside
R1(config-if)#no shutdown
```

```
R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
R1(config)#ip route 0.0.0.0 0.0.0.0 200.200.200.2
R1(config)#access-list 1 permit 192.168.10.0 0.0.0.255
R1(config)#access-list 1 permit 192.168.20.0 0.0.0.255
R1(config)#ip nat inside source list 1 interface gi0/1 overload
R1(config)#
```

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```

R1>enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#ip access-list extended MKT-TO-IT-BLOCK
R1(config-ext-nacl)# deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
R1(config-ext-nacl)# permit ip any any
R1(config-ext-nacl)#exit
R1(config)#
R1(config)#ip access-list extended IT-TO-MKT-BLOCK
R1(config-ext-nacl)# deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
R1(config-ext-nacl)# permit ip any any
R1(config-ext-nacl)#exit
R1(config)#

```

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R-ISP Configuration

```

Router>
Router>enable
Router#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface terminal
      ^
% Invalid input detected at '^' marker.

Router(config)#hostname R-ISP
R-ISP(config)#interface gi0/0
R-ISP(config-if)#ip address 200.200.200.2 255.255.255.252
R-ISP(config-if)#no shutdown

R-ISP(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
R-ISP(config)#
R-ISP(config)#interface gi0/1
R-ISP(config-if)#ip address 8.8.8.1 255.255.255.0
R-ISP(config-if)#no shutdown

R-ISP(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
R-ISP(config)#ip route 192.168.10.0 255.255.255.0 200.200.200.1
R-ISP(config)#ip route 192.168.20.0 255.255.255.0 200.200.200.1
R-ISP(config)#

```

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Extended ACL Configuration on R1

```
R1>enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
R1(config)#ip access-list extended MKT-TO-IT-BLOCK
R1(config-ext-nacl)# deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
R1(config-ext-nacl)# permit ip any any
R1(config-ext-nacl)#exit
R1(config)#
R1(config)#ip access-list extended IT-TO-MKT-BLOCK
R1(config-ext-nacl)# deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
R1(config-ext-nacl)# permit ip any any
R1(config-ext-nacl)#exit
R1(config)#
R1(config)#interface gi0/0.20
R1(config-subif)#ip access-group MKT-TO-IT-BLOCK in
R1(config-subif)#exit
R1(config)#interface gi0/0.10
R1(config-subif)# ip access-group IT-TO-MKT-BLOCK in
R1(config-subif)#exit
R1(config)#
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R1#
```

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Verifying ACL

```
R1#show access-lists
Standard IP access list 1
 10 permit 192.168.10.0 0.0.0.255 (16 match(es))
 20 permit 192.168.20.0 0.0.0.255 (52 match(es))
Extended IP access list MKT-TO-IT-BLOCK
 10 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255 (8 match(es))
 20 permit ip any any (29 match(es))
Extended IP access list IT-TO-MKT-BLOCK
 10 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255 (4 match(es))
 20 permit ip any any (8 match(es))

R1#
```

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Confirms Inbound access list is IT-TO-MKT-BLOCK

```
20 permit ip any any (8 match(es))

R1#show ip interface gi0/0.10
GigabitEthernet0/0.10 is up, line protocol is up (connected)
 Internet address is 192.168.10.1/24
 Broadcast address is 255.255.255.255
 Address determined by setup command
 MTU is 1500 bytes
 Helper address is not set
 Directed broadcast forwarding is disabled
 Outgoing access list is not set
 Inbound access list is IT-TO-MKT-BLOCK
 Proxy ARP is enabled
 Security level is default
 Split horizon is enabled
 ICMP redirects are always sent
 ICMP unreachable are always sent
 ICMP mask replies are never sent
 IP fast switching is disabled
 IP fast switching on the same interface is disabled
 IP Flow switching is disabled
 IP Fast switching turbo vector
 IP multicast fast switching is disabled
 IP multicast distributed fast switching is disabled
 Router Discovery is disabled
 IP output packet accounting is disabled
 IP access violation accounting is disabled
 TCP/IP header compression is disabled
 RTP/IP header compression is disabled
 Probe proxy name replies are disabled
 Policy routing is disabled
 Network address translation is disabled
 BGP Policy Mapping is disabled
 Input features: MCI Check
 WCCP Redirect outbound is disabled
 WCCP Redirect inbound is disabled
 WCCP Redirect exclude is disabled

R1#
```

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Ping Verification for The Topology

1. PC1 to PC2 (in the same VLAN) ping 192.168.10.11 = Success

```
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

2. PC1 to Server0 (IT) ping 192.168.10.20 = Success

```
C:\>ping 192.168.10.20

Pinging 192.168.10.20 with 32 bytes of data:

Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128
Reply from 192.168.10.20: bytes=32 time=27ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 27ms, Average = 6ms

C:\>
```

3. PC1 to Internet ping 8.8.8.8 = Success

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Request timed out.
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

4. PC3 to Internet ping 8.8.8.8 = Success

```
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126
Reply from 8.8.8.8: bytes=32 time<1ms TTL=126

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```


5. PC3 to Server0 (IT) ping 192.168.10.20 = Destination host unreachable

```
C:\>ping 192.168.10.20

Pinging 192.168.10.20 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 192.168.10.20:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

6. PC3 to PC1 (IT) ping 192.168.10.10 = Destination host unreachable

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

7. PC1 to PC3 (Marketing) ping 192.168.20.10 = Destination host unreachable

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Conclusion

Confirmed each of the 3 requirements was met:

- Both VLANs reach internet
- IT can talk to itself, protected from Marketing
- Marketing internet-only, no IT/server access